Capstone Three Project Report

Coral Reef Health From Pollution Indicators

Problem Statement

Coral reefs are among the most at-risk ecosystems, facing increased bleaching and decay. This project will explore the relationship between declining reef health and pollution, using HDI (human development index) and global pollution readings for SO2, NOx, organic carbon, fine particulate matter, and mercury.

Data Cleaning

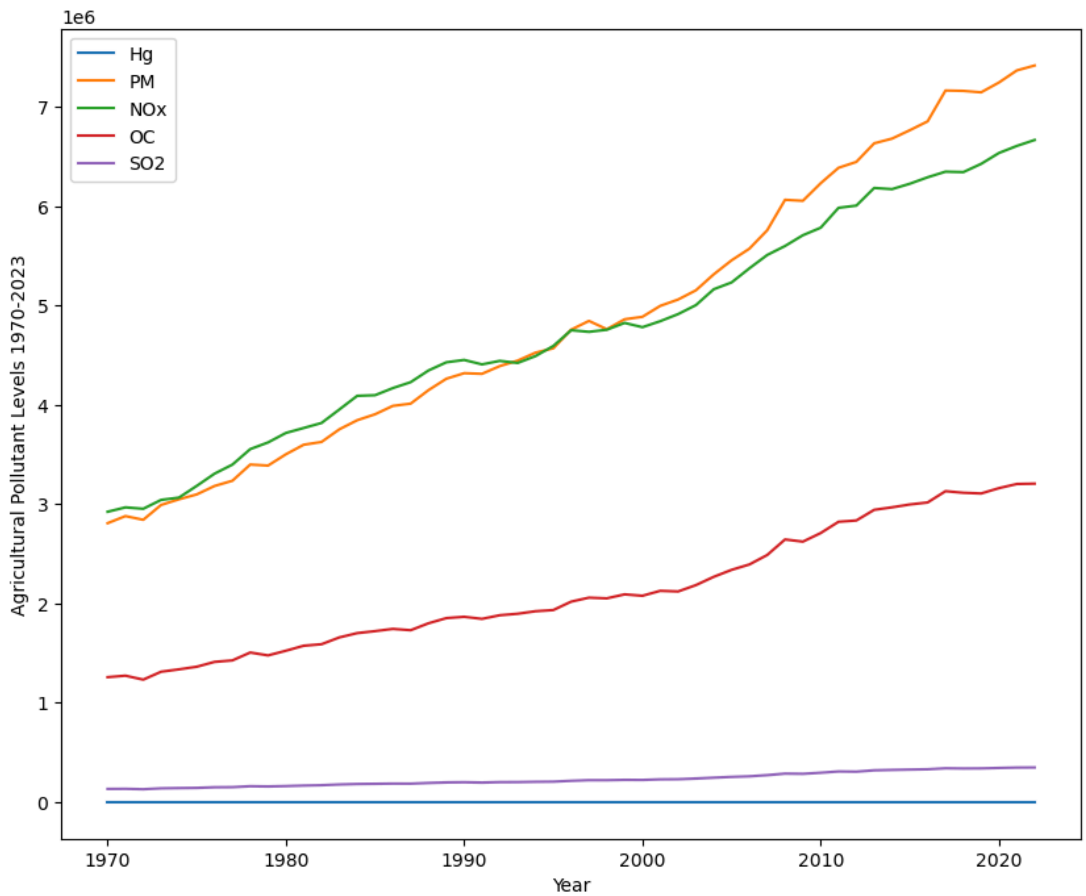
Sources

1. EDGAR
    a. All pollution data was downloaded as an Excel file. The data required reformatting and cleaning. Ultimately, I had to drop 8 columns and 3 rows. This included entries for 271 countries.
2. Coral Reef
    a. The coral reef data initially had 62 columns, but I used only 27 columns. The columns that were kept were key indicators of coral reef health.

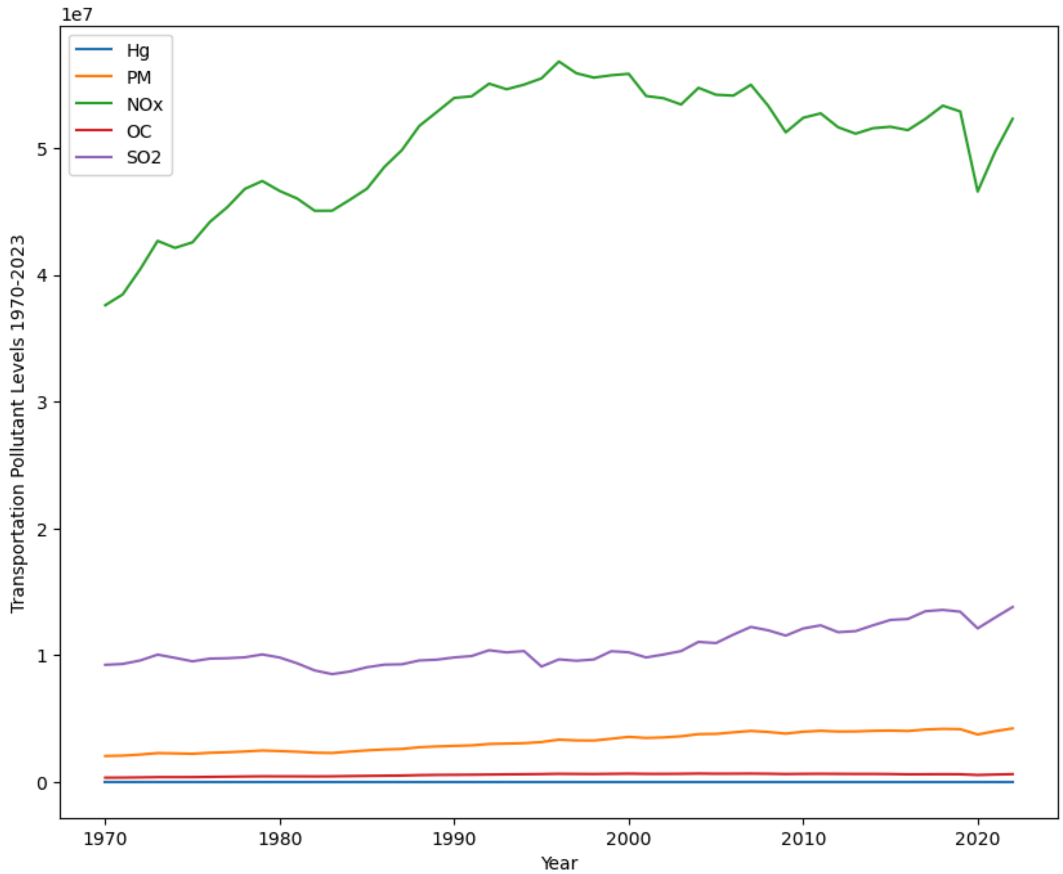
Exploratory Data Analysis

Trends

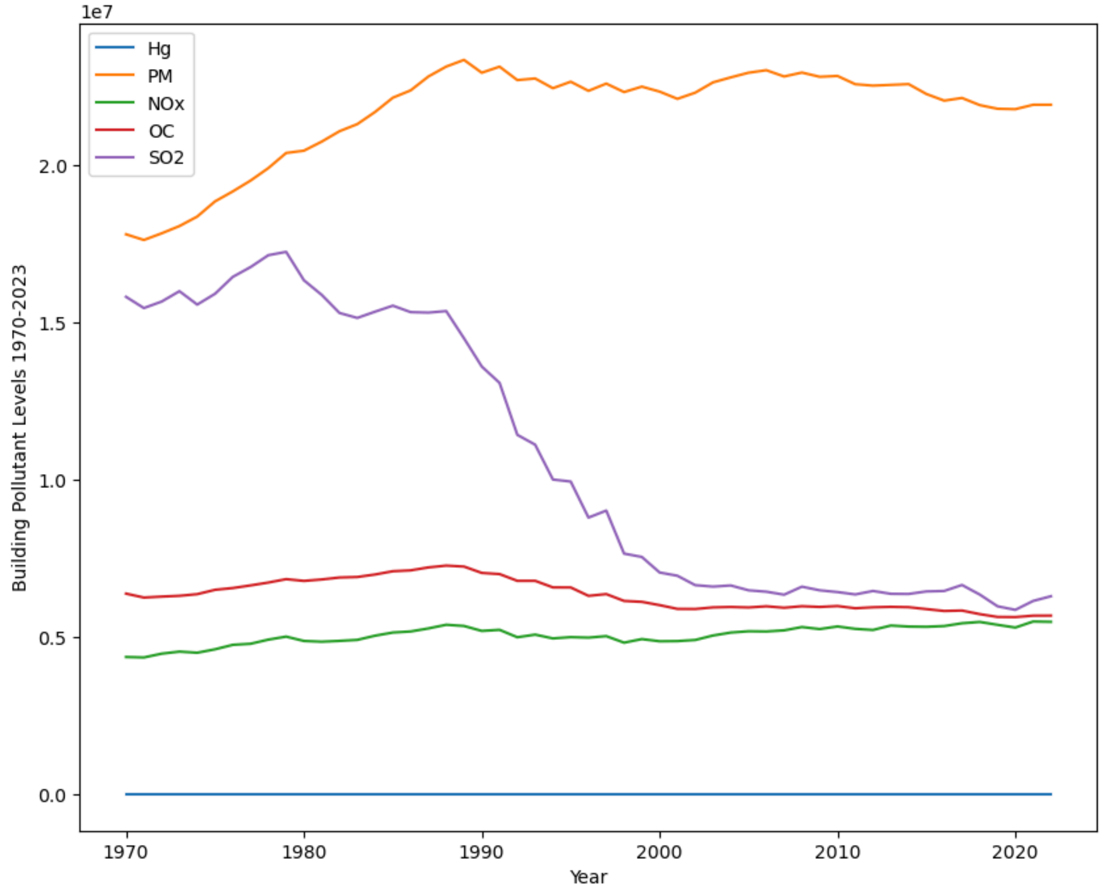
1. Agriculture: Three pollutants have been increasing for the entire time range. Particulate matter is increasing at the highest rate, surpassing NOx. Many aspects of agricultural practices contribute to pollution levels, including livestock emissions and farm equipment emissions, burning, and pesticide use. These pollutants make their way to the ocean and greatly impact ocean life.



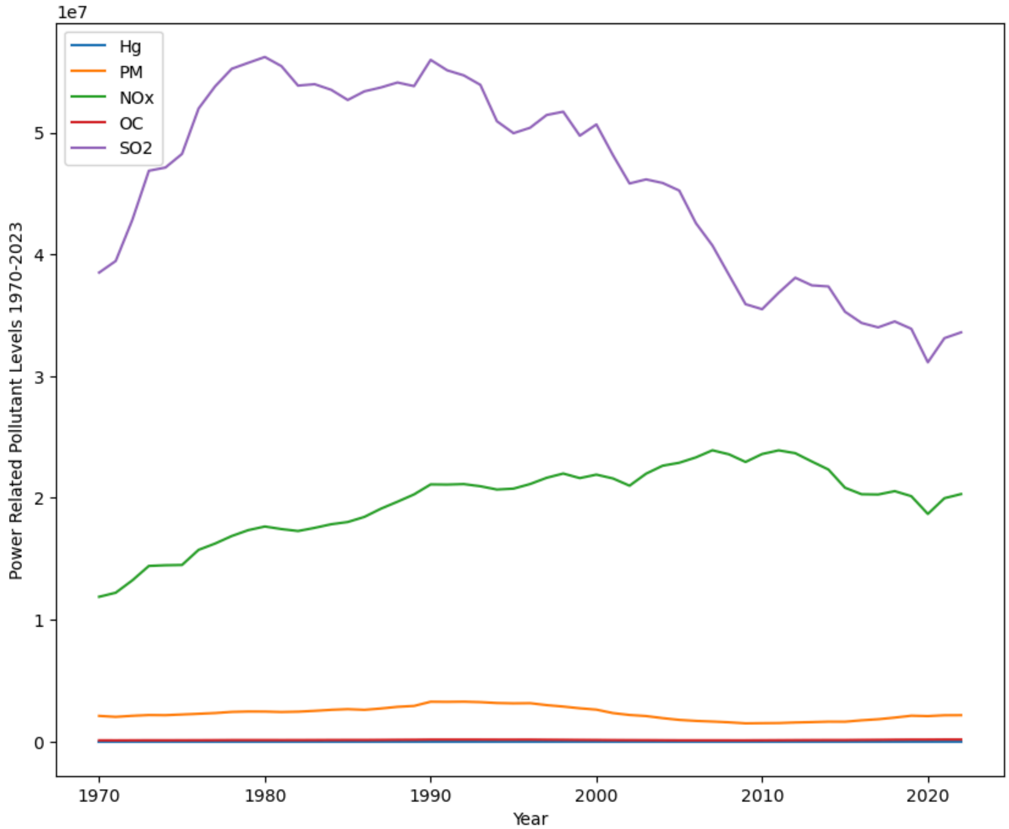
2. Transportation:



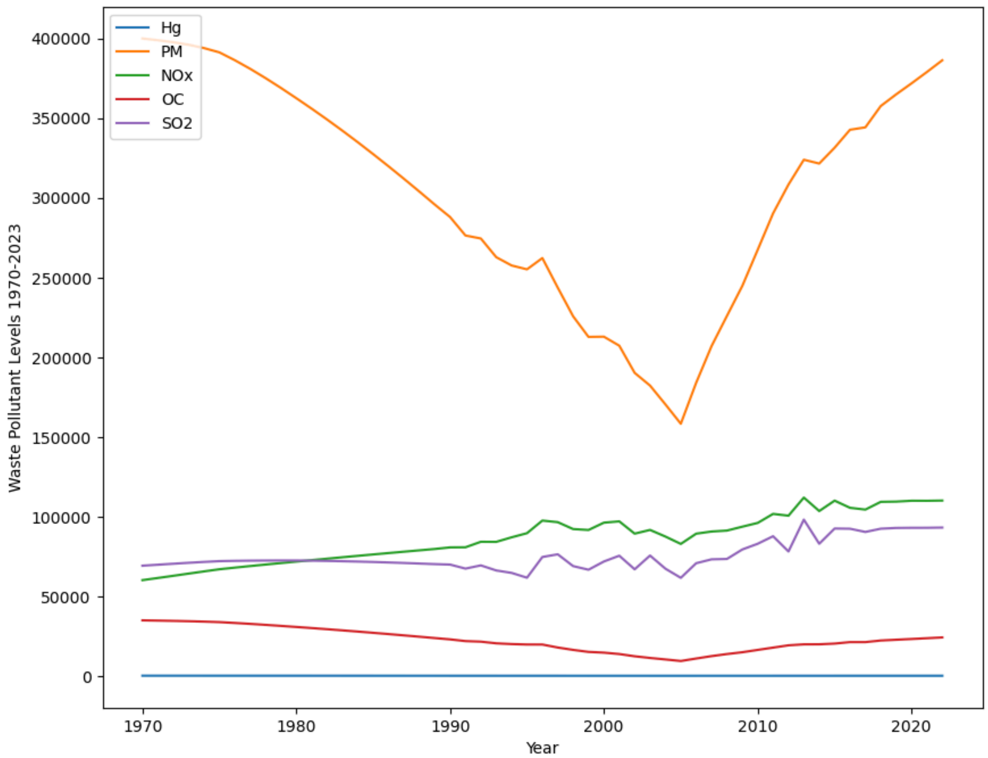
3. Building: Sulfur dioxide levels have decreased substantially since 1980. The other two pollution indicies remained mostly stable, with an slight decline since 1990. Particulate matter, however, has increased.



4. Power Industry: SO2 is all over the place, but thankfully is on a primarily declining trend since 1990. NOx has steadily increased for the entire time range of the data. All other pollutants show lo to no change, strongly suggesting the power industry mainly contributes to SO2 and NOx levels.



5. Waste: There is a very sharp drop and increase in particulate matter in the early 2000s. The 3 other pollutants remain relatively stable, with fluctuations in SO2 and NOx.the sudden drop in particulate matter could possibly be from a false reading, it is worth exploring the reason behind this shift in another project,



Model Selection and Results

Each model is tested using country level data, the HDI information, and global data, pollution levels. The model structure seems to matter more than the data being country or globally scaled. This is demonstrated through similar model metrics between the two model versions and feature importance.

Top Models

	Model	Number	R2	MAE	MSE	RMSE
19	Stacked	Stacked	0.694232	4.632564	107.348315	11.890630
18	Stacked	Stacked	0.689411	4.973147	114.097903	10.681662
21	Random Forest	10	0.688835	4.283365	109.243042	11.890630
13	Random Forest	7	0.688158	4.293881	109.480977	11.890630
20	Random Forest	10	0.681647	4.710488	116.950220	10.814353

Nine distinct model pipelines were used in this project. These include SVR, XGBoost, random forest, linear regression, a stacked model (a model combining MLP regressor, random forest, LGM, XGBoost, with the meta model being linear regression), MLP regressor with two configurations, lightGBM, and ridge regression.

The stacked models performed the best, but only marginally better than random forest when comparing R2 scores. When ordering the results by the lowest (best) RMSE, we see slightly different outcomes. Stacked and random forest are still the highest performing models, with XGBoost joining the top five. RMSE is a good model metric as it penalizes larger mistakes.

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12	Random Forest	7	0.679526	4.723119	117.729227	10.850310
14	XGBoost	8	0.674689	5.373612	119.506137	10.931886
16	LGM	9	0.616868	6.394154	140.747417	11.863702

The table below shows the top five models when ordering them by MAE. This metric scores errors equally and it gives the average size of the prediction errors.

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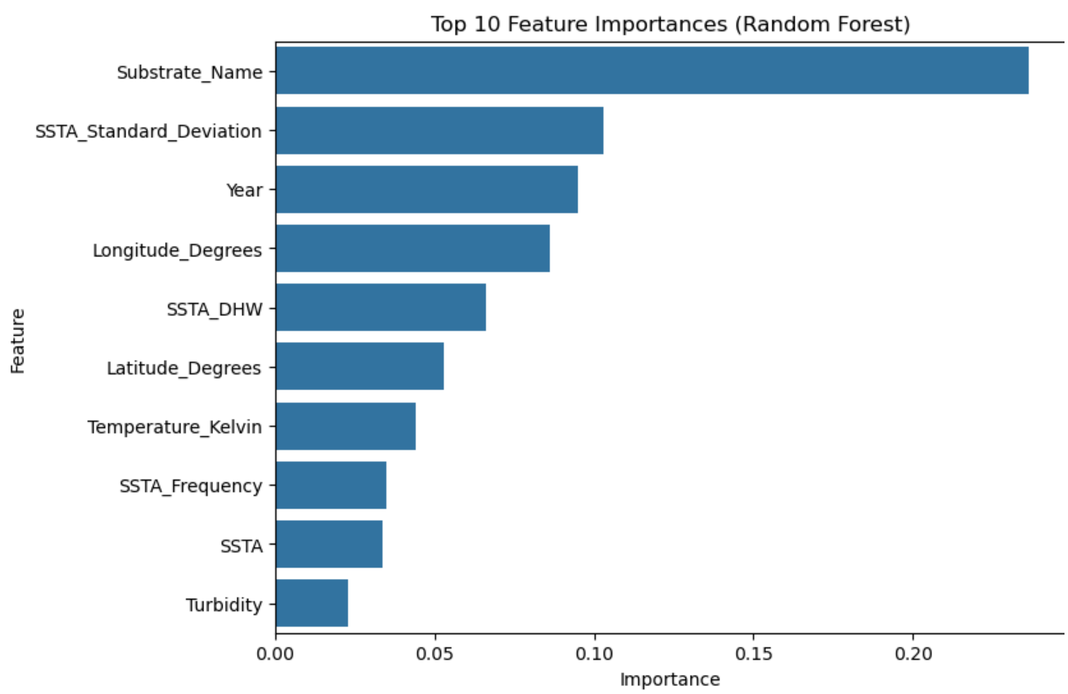
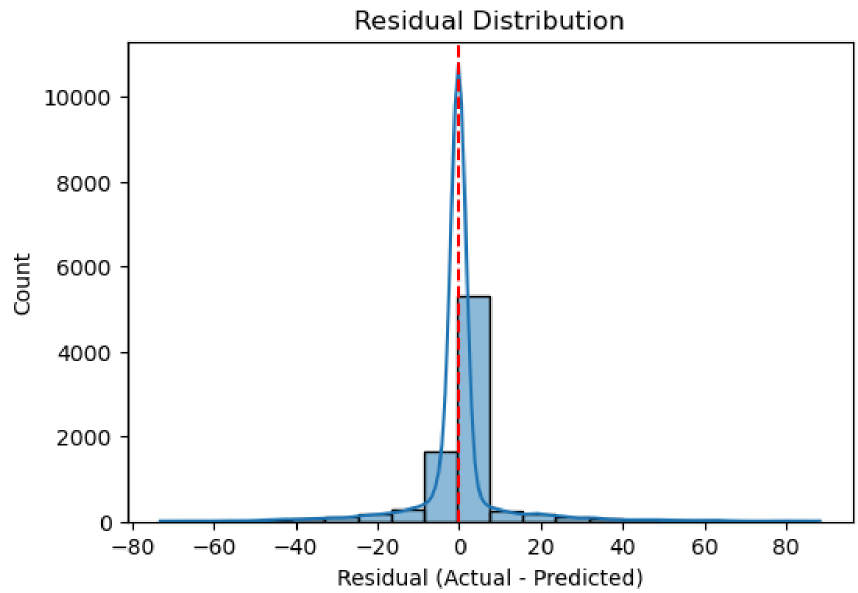


Above are important features for the random forest models. It is interesting to see how the high importance of substrate name. The names were encoded prior to modeling. Year is also an important factor. Other features describe the reef location, how many times the temperature was measured to be outside normal levels, and water quality. This graph specifically shows the important features for the pollution dataset.



The residuals plot looks fairly as most of the residuals are close to 10 or less. Most of the residuals are found to the right of the dotted line, showing the random forest model is under predicting values.

Recommendations

This project met expectations for ecological data predictability with R2 scores of 0.69. The model struggles to deal with outliers and tends to underfit higher bleaching percentages. The important features all being from the coral reef dataset indicate that our pollution and HDI data may not be helpful in predicting bleaching. This should be tested further using different models, as regression models can struggle with outliers.

Future models will likely be stacked, using more robust models that aren't sensitive to outliers. Hyperparameter tuning will be prioritized over testing many model types.